

Representation and Tractability in Set Theoretic Estimation: Sheaves, Rectangles, and the Coupling Obstruction

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Abstract

The feasible set of a set theoretic estimation (STE) problem [2] lives in $\mathcal{P}(\Xi)$, and when the hypothesis space is a product over N variables, $|\Xi|$ is exponential in N : representing the feasible set extensionally hits a 2^N wall that is machine-checked in this project (`Ste.DynamicFrame.Model.card_allExactStates`). This note asks when the feasible set can be represented *locally* and glued, and answers three pieces of the question with Lean-verified results in the module `Ste.Sheaf`. (i) *Gluing on the constraint side*: the assignment $J \mapsto \text{feas}(J)$ from constraint subsets to partial feasible sets sends unions to intersections (`partialFeasibilitySet_iUnion`), so any cover of the constraint index recovers the global feasible set as the limit of local ones (`feasibilitySet_eq_iInter_cover`); on this side there is no obstruction. (ii) *A tractable upper bound*: if every constraint is variable-separable (a rectangle), the feasible set is itself a rectangle (`rectangular_feasibilitySet`) representable by one subset per variable— $\sum$ -space, not $\prod$ -space—with product cardinality (`rectangular_encard`). (iii) *The obstruction*: a single coupling constraint, the diagonal on two bits, is provably not a rectangle (`diagonal_not_rectangular`). This is the elementary form of the Abramsky–Brandenburger contextuality obstruction [1], and the same shape as this project’s non-transitivity countermodel (`Ste.DynamicFrame.Counterexample`). A clearly-labelled outlook sketches the unproven cohomological program: k -consistency as higher gluing, $\tilde{H}^1$ as the measure of intractability, and a treewidth-style tractability dichotomy [3, 4].

1 The problem: the 2^N wall

Combettes’ set theoretic estimation [2] specifies a solution space Ξ , an index set I of pieces of information, and a property set $S_i \subseteq \Xi$ for each $i \in I$; the feasible (solution) set is $S = \bigcap_i S_i$. The framework is silent about *representation*: it treats S as a set, not as a data structure.

The silence becomes a cost the moment Ξ is a product. In the estimation problems this project cares about—dynamic frames over documents, constraint grammars, hyperframe lattices—the hypothesis space factors as $\Xi = \prod_{v \in V} A_v$ over a family of variables (claims, mentions, cells), and $|\Xi| = \prod_v |A_v|$ is exponential in $|V|$. The finite solver in `Ste.DynamicFrame.FiniteSolver` makes the wall precise and machine-checked: over N ambient worlds there are exactly 2^N extensional states (`Model.card_allExactStates`), and crisp STE has full subset expressivity—a single property set can carve out *any* subset of the universe (`arbitrary_subset_as_single_property`)—so without structural assumptions on the constraints, no representation smaller than 2^N can be complete.

The question of this note: *which structural assumptions buy which representations?* The natural language for the question is sheaf-theoretic, because “represent locally and glue” is exactly what a sheaf condition asserts.

2 Gluing over the constraint poset

Fix $S : I \rightarrow \mathcal{P}(\Xi)$ and write, for a subset $J \subseteq I$ of the constraint index,

$$\text{feas}(J) = \bigcap_{i \in J} S_i \quad (\text{partialFeasibilitySet } S \ J),$$

the set of hypotheses consistent with the information in J . The assignment $J \mapsto \text{feas}(J)$ is a presheaf on the poset $\mathcal{P}(I)$ (ordered by inclusion, mapping contravariantly): if $J \subseteq K$ then $\text{feas}(K) \subseteq \text{feas}(J)$, which is the information-monotonicity lemma `partialFeasibilitySet_antitone` already verified in `Ste.Basic`.

The new, machine-checked content of `Ste.Sheaf` is that this presheaf satisfies the gluing condition exactly, with no sheafification needed. The engine is the Formal Concept Analysis bridge of `Ste.HyperFrame` [5]: partial feasibility is definitionally the lower polar of the satisfaction context (`partialFeasibilitySet_eq_lowerPolar`), and polars turn unions into intersections.

Theorem 1 (`partialFeasibilitySet_iUnion`). *For any family $\mathcal{J} : \kappa \rightarrow \mathcal{P}(I)$ of constraint subsets,*

$$\text{feas}\left(\bigcup_k \mathcal{J}_k\right) = \bigcap_k \text{feas}(\mathcal{J}_k).$$

Theorem 2 (`feasibilitySet_eq_iInter_cover`). *If $\bigcup_k \mathcal{J}_k = I$ is any cover of the constraint index, then the global feasible set is the intersection of the local ones:*

$$\text{feas}(I) = \bigcap_k \text{feas}(\mathcal{J}_k).$$

In sheaf language: the functor `feas` sends colimits (unions) of constraint sets to limits (intersections) of feasible sets, so *local consistency data over any cover glues uniquely to the global feasible set*. On the constraint side of the Galois connection there is no obstruction whatsoever—this is a theorem, not a heuristic. Whatever makes STE hard does not live here.

3 The tractable upper bound: rectangular constraints

Where the hardness *can* live is the variable side. Take $\Xi = \prod_{v \in V} A_v$. Call a constraint *rectangular* (variable-separable, a cylinder in every coordinate) if it has the form $\prod_v P_v$ for some family of per-variable sets $P_v \subseteq A_v$ —in Lean, `Set.univ.pi P`. A rectangular constraint inspects each variable independently; it couples nothing.

Theorem 3 (`rectangular_feasibilitySet`). *If every constraint is rectangular, $S_i = \prod_v P_{i,v}$, then the feasible set is itself the rectangle whose v -th side is the pointwise intersection:*

$$\bigcap_i \prod_v P_{i,v} = \prod_v \left(\bigcap_i P_{i,v} \right).$$

This is the representation theorem for the easy case: a fully variable-separable STE problem is represented *exactly* by one subset per variable. The representation lives in $\sum_v |A_v|$ space—linear in the number of variables—never in the $\prod_v |A_v|$ product space. The 2^N wall does not exist for rectangular constraint families.

The cardinality statement is checked as well, in terms of Mathlib’s extended natural cardinality `encard` (which handles infinite sides gracefully):

Theorem 4 (`rectangular_encard`). *For a finite variable set V (`Fintype V`),*

$$\left| \bigcap_i \prod_v P_{i,v} \right| = \prod_{v \in V} \left| \bigcap_i P_{i,v} \right|,$$

as extended cardinals.

So in the separable case even the *size* of the feasible set is a product of locally-computable sizes: consistency checking, counting, and sampling all decompose per-variable. This is the STE incarnation of the trivial end of CSP tractability: a constraint network with no edges is backtrack-free [4].

4 The obstruction: coupling is not rectangular

The lower bound is a single, minimal counterexample. Let the hypothesis space be two binary variables, $\Xi = (\text{Fin } 2 \rightarrow \text{Bool})$, and consider the *diagonal* (coupling, equality) constraint

$$\Delta = \{ f \mid f(0) = f(1) \} \quad (\text{diagonal}).$$

Theorem 5 (`diagonal_not_rectangular`). *There is no family $t : \text{Fin } 2 \rightarrow \mathcal{P}(\text{Bool})$ with $\Delta = \prod_v t_v$.*

The proof is the two-line contextuality argument: $(0, 0) \in \Delta$ and $(1, 1) \in \Delta$, so any rectangle containing Δ has $0 \in t_0$ and $1 \in t_1$, hence contains the mixed point $(0, 1)$ —which violates the coupling. The local (per-variable) data of Δ is *full*: projected to either coordinate, everything is possible. The constraint is invisible locally and exists only jointly.

This is precisely the elementary form of the sheaf-theoretic obstruction of Abramsky and Brandenburger [1]: a family of perfectly consistent local sections (here: both values possible at each variable) that admits no factorization through a product—the “global section” data is not determined by, and not recoverable from, the per-variable marginals. In cohomological terms (made precise only in the outlook below), the diagonal carries a nonzero Čech gluing class relative to the cover of V by singletons.

It is also, structurally, the same countermodel this project already uses for a different purpose: `Ste.DynamicFrame.Counterexample` refutes transitivity of possible-coreference (`maySame_not_transitive`) with a three-world model in which pairwise-consistent local identifications fail to assemble into a global one. Pairwise consistency without global consistency is the one phenomenon, seen twice.

Consequences for representation: since rectangles are closed under intersection (`rectangular_feasibilitySet` again), no family of rectangular constraints can ever express Δ . Coupling constraints are therefore *exactly* the constraints that force representations beyond $\sum$ -space; the 2^N wall of `card_allExactStates` is built out of them.

5 Outlook: the cohomological program (not proven)

Everything in this section is conjecture or programme. None of it is machine-checked, and none of it should be cited as a result of this project.

Conjecture (outlook) (Feasibility as a sheaf on a site over variables). *Section 2 proved gluing over the constraint poset. The substantive open direction is the variable side: assign to each subset $W \subseteq V$ of variables the set of partial assignments on W consistent with all constraints whose*

support lies in W . We conjecture this is a presheaf on a Grothendieck site over $\mathcal{P}(V)$ whose sheaf condition, relative to a cover of V by constraint supports, holds precisely for the tractable fragments; the diagonal example shows the presheaf is not a sheaf for the singleton cover in general.

Conjecture (outlook) (k -consistency is higher gluing). *Local (k -)consistency in the CSP sense [3, 4]—every consistent assignment on $k - 1$ variables extends to any k -th—should be exactly the statement that sections glue along the cover of V by k -element subsets. Freuder’s theorem (strong k -consistency plus width $< k$ implies backtrack-free search) would then become: on covers refined enough for the treewidth, the presheaf is a sheaf, and a global section is constructed greedily.*

Conjecture (outlook) ($\check{H}^1$ measures intractability). *Following Abramsky and Brandenburger [1], the obstruction to a family of compatible local sections extending to a global one should be captured by a Čech cohomology class in $\check{H}^1$ of the support cover with coefficients in the (relative) presheaf of partial solutions; the class vanishes iff gluing succeeds. The conjectured quantitative version: the difficulty of an STE instance—the size blow-up of its smallest exact representation—is controlled by the size of this obstruction, with rectangular families the $\check{H}^1 = 0$ case of Section 3 and the diagonal of Section 4 the minimal nonvanishing class.*

Conjecture (outlook) (Tractability dichotomy). *Bounded treewidth of the constraint (co)occurrence structure—or, conjecturally equivalent in this setting, bounded obstruction in the sense above—yields a polynomial-size glued representation (junction-tree style: rectangles over bags, glued along separators), and unbounded coupling reproduces the 2^N wall. This would be the STE form of the classical CSP dichotomy between bounded-width tractability and general intractability [3].*

What stands *proven* against this programme are its two endpoints: the fully separable case is exactly linear (`rectangular_feasibilitySet`, `rectangular_encard`), and one coupling suffices to leave it (`diagonal_not_rectangular`)—with the constraint-side gluing (`feasibilitySet_eq_iInter_cover`) guaranteeing that the difficulty is genuinely about variables, not about how information is indexed.

References

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